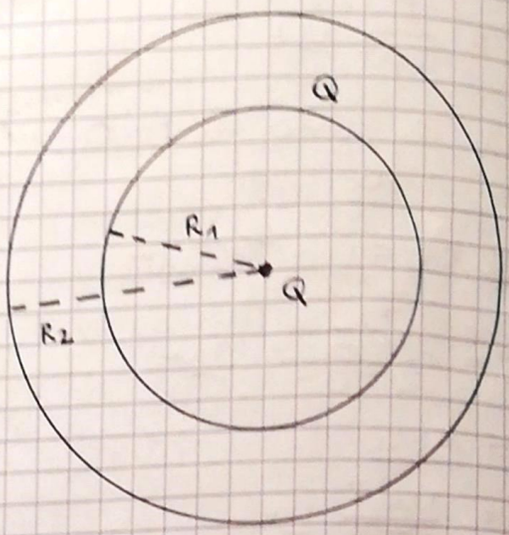
APPELLO 14/06/2019

ESERCIZIO 1



SE $0 < r \leq R_1$

$$\underline{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\,\underline{u}_r$$

SE $R_1 < r < R_2$

$$\underline{E} = 0$$

SE $r > R_2$

$$\underline{E} = \frac{Q}{2\pi\varepsilon_0 r^2}\,\underline{u}_r$$

POTENZIALE ($V_\infty = 0$)

$$V(r) - V_\infty = -\int_\infty^r \frac{Q}{2\pi\varepsilon_0 r^2}\,dr = \frac{Q}{2\pi\varepsilon_0 r}$$

SE $R_1 < r < R_2$

$$V(r) - V(R_2) = 0 \Rightarrow V(r) = V(R_2) = \frac{Q}{\pi\varepsilon_0 R_2}$$

SE $0 < r < R_1$

$$V(r) - V(R_1) = -\int_{R_1}^r \frac{Q}{4\pi\varepsilon_0 r^2}\,dr = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r} - \frac{1}{R_1}\right)$$

$$\Rightarrow V(r) = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r} - \frac{1}{R_1} + \frac{2}{R_2}\right)$$

Se il guscio viene collegato a terra la sfera esterna si carica negativamente di carica $-Q$.

Dunque:

se $r > R_1 \Rightarrow \underline{E} = 0$

se $0 < r \leq R_1 \Rightarrow \underline{E} = \frac{Q}{4\pi\varepsilon_0 r^2}\,\underline{u}_r$

Da cui

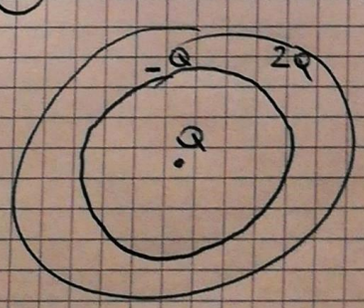
SE $r > R_1$

$$V(r) - \underbrace{V(R_1)}_{=0} = -\cancel{\int_{R_1}^r} 0$$

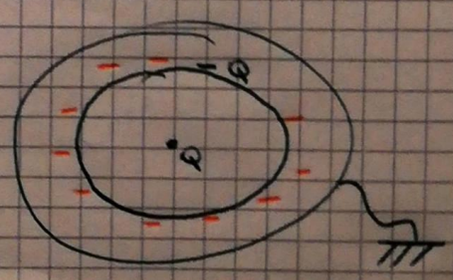
SE $0 < r \leq R_1$

$$V(r) - \underbrace{V(R_1)}_{=0} = -\int_{R_1}^r \frac{Q}{4\pi\varepsilon_0 r^2}\,dr = \frac{Q}{4\pi\varepsilon_0}\left(\frac{1}{r} - \frac{1}{R_1}\right)$$

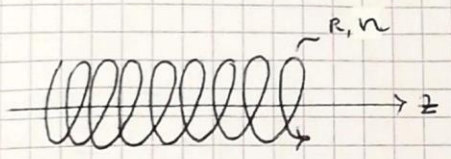
(a)



(b)



ESERCIZIO 3



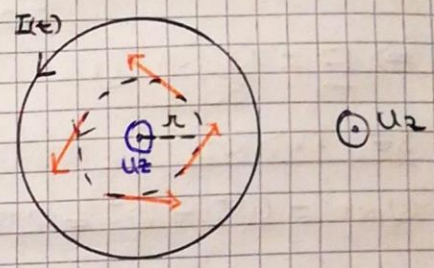
$I(t) = -\frac{i_0}{T}t + i_0$

$\underline{B}(t) = \mu_0 n I(t) \underline{u}_z$ dentro il solenoide

$\underline{B}_f(t) = 0$ fuori dal solenoide

So anche che

$f = \oint_\gamma \underline{E} \cdot \underline{u}_t \, dl = -\frac{d\Phi(B)}{dt}$



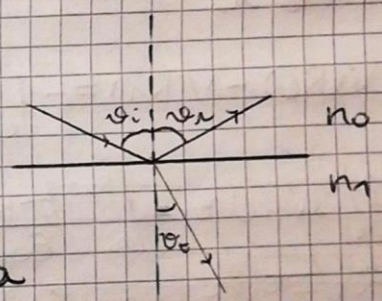
$\Phi(\underline{B}) = \pi R^2 B = \pi R^2 \mu_0 n \left(i_0 - \frac{i_0}{T}t\right)$

$\Rightarrow 2\cancel{\pi} r E = \cancel{\pi} R^2 \mu_0 n \frac{i_0}{T}$

$\underline{E} = \frac{\mu_0 n R^2 i_0}{2 T r} \underline{u}_t$

ESERCIZIO 4

$\vartheta_i = 60°$



$tg\,\vartheta_i = \frac{n_1}{n_0}$

$\Rightarrow n_1 = tg\,\vartheta_i = \sqrt{3}$

la luce riflessa è polarizzata linearmente con direzione di polarizzazione perpendicolare al piano di incidenza.

$R = 50\,\%$